

The Semi-Humanoid Robot Market 2026

Global competitive landscape: wheeled, tracked and pedestal dual-arm humanoid robots

217 robots · ~200 companies · China · USA/Canada · Europe · Japan · Korea · Rest of world

Market study compiled 2 August 2026

Roboteur Research

Andreas Köpf (x.com/neurosp1ke)

compiled by Claude Fable 5

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1 Executive summary

The semi-humanoid — a human-like dual-arm upper body on a wheeled, tracked or pedestal base — is where humanoid robotics is actually being deployed in 2026. While bipedal humanoids dominate headlines and venture valuations, wheeled dual-arm robots dominate paying work: of the 217 systems documented in this study, 82 are shipping products and a further 31 are formally announced. China's "2026 Wheeled Humanoid Robot Industry Development Blue Book" counts >5,000 wheeled-humanoid units shipped in China in 2025 (~25–28% of Chinese humanoid volume), and Interact Analysis states plainly that wheeled platforms are "the preferred near-term form factor for real-world industrial deployment." Morgan Stanley's influential humanoid forecasts *exclude* wheeled-base torsos entirely — meaning the segment this study covers is systematically under-represented in headline market numbers, and correspondingly under-analyzed by competitors.

Seven findings that shape the opportunity:

- **China leads on volume and velocity.** 63 of 217 documented robots are Chinese; AgiBot alone shipped 5,100+ units in 2025 and passed 15,000 cumulative robots by June 2026. Real factory orders exist (Fulin Precision ~100× A2-W; Longcheer's hundreds-of-millions-RMB G2 order), and the supplier layer (chassis, actuators, hands) is commoditizing fast — entry prices now reach ¥26,900 (Unitree R1-D) and ~\$1,000–1,500 (SwitchBot Onero H1 target).
- **The US pivoted from bipeds to economics.** The 2025–26 US wave (Sunday Memo, Weave Isaac at \$7,999, Dexterity Mech, Dyna, Walden Robotics at a \$1.1B valuation out of stealth with Toyota as investor-customer) sells labor economics — RaaS by the hour, home robots by subscription — not anthropomorphism.
- **Regulation just redrew the map.** FCC Public Notice DA 26-786 (28 July 2026) added foreign-produced "advanced robotic devices" to the US Covered List: new foreign-made mobile, networked, AI-enabled robots — wheeled semi-humanoids squarely included — lose US equipment authorization unless granted a conditional approval. This fences the largest single market against Chinese, Korean, Japanese *and* European vendors alike, and materially raises the value of domestic/allied manufacturing footprints.
- **Europe is strong in research, fragmented in product.** PAL Robotics, Pollen/ Hugging Face, Neura's MiPA (€9,999 announced, deliveries slipping), igus's €47,999 Iggy Rob, Humanoid UK's \$1.35B valuation and Duatic's ETH-bred Alpha coexist with the cautionary tales of Aldebaran/Pepper (liquidated 2025) and Cybedroid. German research depth (DLR, KIT, TUM, FZI, Fraunhofer lineage) is world-class and under-commercialized — a talent and IP pool a German entrant can draw on.
- **Incumbents are converging on exactly this form factor.** Toyota's ELEY (QDD joints, scapular axis, wheeled base), Yaskawa's MOTOMAN NEXT + its quiet ownership of Tokyo Robotics, Midea's six-armed MIRO U, Kawasaki/Foxconn's Nurabot, Hexagon's AEON, plus component giants (Schaeffler backing Humanoid UK, Hexagon and reportedly Neura; LG investing in AgiBot and Dexmate while building its own CLOiD).
- **Data is the second product.** Nearly every serious player monetizes demonstration data: AgiBot World's million-trajectory dataset, Sunday's 2,000 paid glove-wearing "memory developers", 1X's security fleet feeding NEO, China's 40+ state-funded training centers. A hardware entrant without a data flywheel strategy competes on margins; with one, on compounding capability.
- **Certification favors wheels — for now.** Statically stable platforms certify under existing frameworks (ISO 13482, ISO 10218:2025, ANSI R15.08, UL 3300) while dynamically balancing robots await ISO 25785-1 (2026–27). The EU Machinery Regulation 2023/1230 (applies 20 Jan 2027, with AI-behaviour and cybersecurity requirements) is an opportunity for an EU vendor to lead on compliance-by-design.

Bottom line for a German entrant: the semi-humanoid category is real, growing, and — outside China — not yet locked up. The winning wedge combines (1) an industrial or care use case with demonstrable ROI, (2) compliance-by-design for the 2027 EU regime, (3) a data-collection strategy from day one, and (4) supply-chain positioning that exploits the new US restrictions on Chinese hardware rather than suffering them. Section 16 details the white spaces.

2 Scope, definition & methodology

Definition

A **semi-humanoid robot** has a human-like upper body — head and/or torso with **two arms** — whose legs are replaced by another locomotion or mounting concept: a wheeled base (differential, omnidirectional/mecanum, swerve, two-wheel balancing or ball-balancing), a tracked base, a lift column, or a fixed pedestal. SoftBank's Pepper is the textbook example; AgiBot's A2-W, Galbot's G1, Rainbow's RB-Y1 and Sunday's Memo are the 2024–26 generation. Included statuses: shipping, announced, prototype, research and notable discontinued systems. Excluded: bipedal humanoids, single-arm mobile manipulators (documented in Annex A where influential — Toyota HSR, Hello Robot Stretch), arm-only cobots, armless telepresence robots, and dual-arm machines without an anthropomorphic upper body. Wheel-leg hybrids (Tencent The Five, GAC GoMate, Hexagon AEON, Moon Dynamics L1) sit in the annex as borderline.

Method & data quality

Research was conducted in August 2026 through multi-lingual web research (English, Chinese, Japanese, Korean, German, French, Spanish, Russian): official vendor documentation and datasheets, trade press, financial media and filings, academic papers, community databases, teardown/forum material, and curated robot-description repositories. Every robot has a dossier (company profile, technical dossier, machine-readable spec sheet, source-attributed imagery — all images individually visually verified). **Every spec field carries a per-field source URL and one of three confidence ratings:** *vendor-claimed* (manufacturer statement), *third-party* (independently reported/verified), *estimated* (inferred). The companion xlsx preserves this per-field register (~3,400 source rows); the color coding in its matrix mirrors these ratings. Conflicting figures are retained with both values flagged rather than silently resolved. Where a vendor discloses nothing (a frequent Chinese and stealth-startup pattern), fields read "n/a (not disclosed)" — absence of a number is itself signal.

Caveats. This market moves at weeks-scale: 2026 alone brought the FCC action, Mobileye's \$900M Mentee acquisition, Walden's \$1.1B debut and several WAIC launches. Chinese shipment claims are frequently sell-in, not deployment (Interact Analysis: only ~10% of 2025 humanoid production reached real applications). Analyst forecasts diverge by >25× for 2035; we present ranges, not point estimates.

Corpus at a glance

Region	China	USA	Europe	Japan	Other	Korea	Canada
Robots	60	40	36	31	21	15	3

Status	shipping	prototype	discontinued	announced	research
Robots	82	38	37	31	18

Main-matrix corpus: 206 robots (plus 11 borderline/excluded in Annex A). "Other" combines Israel, India, Russia, Turkey, the Middle East, SE Asia and global/open-source projects. Canada is broken out in the xlsx but merged into the USA & Canada chapter here.

3 What the world calls this category

No single global term has settled; procurement, patent and market searches must cover all of the following. Chinese usage is the most developed — and China's Mobile Robot (AGV/AMR) Industry Alliance published a dedicated wheeled-humanoid Blue Book in May 2026 precisely because "the industry lacks unified standards and consensus."

Language	Term	Literal meaning	Prevalence / notes
Chinese	轮式人形机器人	wheeled humanoid robot	Dominant. Used by UBTech, CASBOT, Topstar, Robot Era; industry analyses claim wheeled models are >80% of deployed "humanoids"
	半人形机器人	semi-humanoid robot	Popularized by Pudu's D7 marketing; common in media
	轮式双臂机器人	wheeled dual-arm robot	Engineering/industrial framing (Fourier GRW, Rokae Helios)
	轮臂式机器人	wheel-arm robot	Emerging (Quectel, Zerith, KNEWBOTS)
	复合机器人	composite robot	Legacy term for arm-on-AMR; adjacent category
	超人形机器人	super-humanoid	New coinage for multi-arm forms (Midea MIRO U six-arm, iSageBot four-arm)
Japanese	セミヒューマノイド	semi-humanoid	Growing standard in 2025–26 trade press (AGIRobots, Pudu coverage); replacing older 上体型
	車輪型ヒューマノイド	wheel-type humanoid	Standard trade-press term, esp. for Chinese imports
	双腕ロボット	dual-arm robot	Dominant industrial term (Yaskawa SDA, Kawasaki); mobility-agnostic
	人型重機 / アバターロボット	humanoid heavy machinery / avatar robot	Niche coinages: Jinki Ittai's railway robot; teleop category (ugo, Ory's 分身ロボット "alter-ego robot")
Korean	이동형 양팔로봇	mobile dual-arm robot	Rainbow Robotics' official RB-Y1 category
	휠 휴머노이드 / 세미 휴머노이드	wheel humanoid / semi-humanoid	Trade press; Seoul Economic Daily's March 2026 "wheeled semi-humanoids gain traction" wave coverage
German	semi-humanoider Roboter / humanoider Roboter auf Rädern	—	Media descriptions; also "mobiler Manipulator", "Pflege-/Assistenzroboter" (application terms), "kognitiver Roboter" (Neura/Oversonic marketing)
	Roboter-Avatar	robot avatar	Telepresence subsegment (Devanthro Robody)
French	robot semi-humanoïde / humanoïde à roues	—	Also "robot majordome" (butler), "robot d'accueil" (reception)
Spanish	robot semihumanoide / humanoide con ruedas	—	Also "manipulador móvil"
English	wheeled humanoid / semi-humanoid / dual-arm mobile manipulator	—	All three common; analysts add "mobile manipulation platform"; Morgan Stanley excludes wheeled torsos from "humanoid" entirely

4 Market size & forecasts

The headline numbers — and why they mislead

Humanoid-robot forecasts span an extraordinary range: for 2035 alone, published market sizes run from ~\$9B (Precedence) through Goldman Sachs' \$38B TAM to ~\$250B (Future Market Insights) — a >25× spread driven by definitions, and long-horizon calls reach \$5T by 2050 (Morgan Stanley) and \$7T (Citi). Two structural facts matter more than any point estimate:

- **Semi-humanoids are under-counted.** Morgan Stanley's definition explicitly excludes "a humanoid torso mounted to a wheeled AMR and other bases that remain stable when powered down." The deployable end of the market is thus largely invisible in the most-quoted numbers — upside for anyone who counts properly.
- **Specialists are far more conservative than banks.** Interact Analysis (May 2026): >700k humanoid units/yr and ~\$15B revenue by 2035, commercial-viability inflection ~2032; 2025 production >20,000 units (10× 2024) but only ~10% deployed in real applications; China took >90% of production. Bank of America: 18k units 2025E → 1M/yr by 2030; BOM ~\$35k end-2025 falling to \$13–17k by 2030–35.

What is actually known about the wheeled segment

- **China Blue Book (AGV/AMR Industry Alliance, May 2026):** China wheeled-humanoid shipments >5,000 units in 2025, market value >RMB 1.8B (~\$250M) — the only hard unit datapoint for the segment; implies wheeled ≈25–28% of Chinese humanoid volume [third-party; derivation estimated].
- **2025 humanoid actuals (all form factors):** Omdia counts 13,318 global shipments with Chinese vendors in the top six (AgiBot 5,100+, Unitree ~4,200–5,500, UBTech ~1,000); Xinhua/CSJ report ~18,000 China-only (+650% YoY). Counting rules differ; both are directionally consistent: China ≈90% of world volume.
- **2026 outlook:** GGII forecasts 62,500 China humanoid shipments; Morgan Stanley doubled its 2026 China forecast to 50,000 units in June 2026; TrendForce sees >700% YoY global growth. Chinese executives talk "100–200k level". Manufacturing deployments are described by China Securities Journal as "essentially all wheeled + dual-arm configurations, because ROI is decisive."
- **Commercial-report shares conflict irreconcilably** (wheeled = 40.6% GMI vs 85.3% Roots vs ~37% FMI) — use unit-level evidence, not percentage claims.

Capital formation

Humanoid startups raised **\$6.1B across 139 deals in 2025** (PitchBook, +300% YoY) and **\$8.6B in the first seven months of 2026** — 1.8× all of 2025. China alone: ~RMB 39.8B across 325 deals in 2025 (+326%), and ≥RMB 100B invested in 2026 YTD — more than the previous five years combined — atop a ~RMB 1T (~\$138B) 20-year state fund and >RMB 70B of municipal funds. Semi-humanoid-relevant rounds include Galbot (~\$1.15B lifetime, \$3B valuation, backers from CATL to China's IC "Big Fund"), AgiBot (Tencent, BYD, LG; backdoor-listed via Shangwei), AI² Robotics (~\$735M at ~\$2.8–3B, July 2026), Atribot (unicorn, June 2026), Sunday (\$1.15B), Walden (\$300M seed at \$1.1B), Humanoid UK (\$1.35B), Dyna (~\$600M), Neura (Series C up to \$1.4B, Amazon/Nvidia participation, confirmed June 2026), and Mobileye's \$900M acquisition of Mentee. Government pull is Chinese policy: MIIT's 2023 guideline targets world-leading humanoid strength by 2027, with a reported 100,000-robot deployment goal.

5 Why wheels: the form-factor debate

The engineering case for replacing legs is quantitative, and vendors increasingly make it openly:

- **Cost.** Galbot founder Wang He: a wheeled chassis is ~10× cheaper than legs and more stable; bipedal locomotion "unnecessary... would increase costs." Wheeled entry prices (\$4,290 Unitree R1-D; \$7,999 Weave Isaac; ~\$14k Astribot T1) sit an order of magnitude below bipedal industrial pilots.
- **Energy.** Legged electric locomotion suffers Joule-heating losses at low speed/high torque. Typical bipeds run 2–4 h (Optimus V2 ~2.3 kWh ≈2 h; Unitree H1 864 Wh <4 h static); wheeled semi-humanoids run 5–16 h and increasingly hot-swap (AgiBot A2-W: 2 kWh, 5 h, swappable; Galbot G1: 8 h; Dexmate Vega: 10–30 h claimed).
- **Safety physics & certification.** Rodney Brooks: falling legs carry high kinetic energy — his prediction is that "humanoid robots will get wheels for feet." A statically stable base stops rather than falls on power loss, and certifies under existing standards today (see §14). Korea's March-2026 wave (T-Robotics: the wheeled hybrid is our "final form... will never build legs") and the UK's Humanoid ("wheels first") made this an explicit strategy, not a compromise.
- **Engineering focus.** Budget not spent on balance goes into arms, hands and autonomy — the parts that earn revenue (1X's EVE-before-NEO data strategy; Dexterity's 60 kg-lift Mech).

The counter-case remains real: stairs, curbs and cable-strewn floors; legs as a positioning system (Agility); the "little teapot" instability of a static base reaching far under load (Melonee Wise); and household TAM arguments that pushed 1X from wheeled EVE to legged NEO. The pragmatic industry answer is increasingly modular lower bodies (Appttronik Apollo wheeled/legged variants; AgiBot X2-N leg-wheel switching) and wheel-leg hybrids (Hexagon AEON, GAC GoMate) — hedges that concede the near-term economy of wheels.

6 Regional landscape — China

63 documented robots — the deepest, fastest, cheapest market. Three strata: (1) **scale leaders** — AgiBot (5,100+ units 2025, 15,000 cumulative by June 2026, Fulin and Longcheer factory orders, backdoor-listed), Galbot (\$3B valuation; licensed robot pharmacies, ~170 unmanned kiosks, factory pilots at Mercedes-Benz China/CATL/ Zeekr), UBTech (Cruzr S2, exported to Japan via GA Robotics), Pudu and Keenon riding 100k/80k-unit service-robot install bases; (2) **technology specialists** — Astribot (10 m/s end-effector speed, \$14k T1), PaXini (~2,000 tactile sensors), Spirit AI (full force control), DexForce (Sim2Real), Robot Era (Haier's "HIVA" is its rebadged Q5); (3) **incumbent entrants** — Midea (six-armed MIRO U in its own washing-machine plant; owns KUKA), JAKA/Rokae/Elite/Topstar/Siasun from the cobot/industrial world, Fourier from rehab, plus ecosystem oddities from four-armed iSageBot to ByteDance's research ByteMini. A dedicated supplier layer (SEER's C1-D humanoid chassis, Realman's joint modules with claimed 50,000 h MTBF, Inspire hands) is commoditizing construction; data-collection platforms (ARX, AgileX, Galaxea, HEXFELLOW) arm the world's embodied-AI labs. Weaknesses: near-zero published safety certifications, sell-in vs deployment gaps, financial distress cases (CloudMinds), and now the US FCC wall.

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
AlphaBot 2 爱宝 (Aibao) AI² Robotics	shipping	wheeled mobile base with retractable waist-leg torso	10	n/a	Geely Jingneng Microelectronics semiconductor fab (03/2025-); Bloomage Biotech cleanrooms; automotive plants;
LIFT2 (ARX-LIFT2) ARX-LIFT2 ARX Robotics	shipping	omnidirectional wheeled base with motorized lift column	5	from CNY 249,800	wide adoption in Chinese research labs as ALOHA-style platform; no unit counts disclosed
A2-W 远征A2-W AgiBot (Zhiyuan Robotics)	shipping	wheeled (4WD omni, zero turning radius, crab-walk)	5	RMB 400k-800k (~USD 56k-112k) insider range, not officially published	Fulin Precision ~100-unit order 08/2025 (tens of millions RMB); 4-robot demo 800+ totes/3h zero failures; batt
Genie G2 精灵G2 AgiBot (Zhiyuan Robotics)	shipping	wheeled (4-DoF omnidirectional chassis: omni, crab-walk, in-	5	not disclosed; predecessor Genie G1 at RMB 450k (~USD 62k)	Longcheer ~1,000-unit framework order (hundreds of millions RMB) for tablet lines; Junpu Intelligent automotiv
Cobot Magic Cobot Magic (Mobile ALOH AgileX Robotics)	shipping	Tracer two-wheel differential AGV base, 685x570x155mm, 100kg	1.5	~USD 48k class via distributors (India listing ₹3,999,999); config-dep	global research-lab adoption; AgileX Tracer base underpinned Stanford Mobile ALOHA
S1 Astribot S1 Astribot (Stardust Intelligence)	shipping	wheeled (3-DoF omnidirectional base) + flexing torso column;	10	research edition ~RMB 500k (~USD 69k); B2B quote; T1 sibling from RMB	JD.com, CCTV, elder-care institutions, retail in six cities; reported thousand-unit Thundersoft order (unverif
T1 Astribot T1 Astribot / Stardust Intelligence	shipping	wheeled (compact base)	5	from RMB 89,900 (~USD 13k); higher-spec variant ~RMB 99,000 reported	first unit shipped 2026-06-01; mass production reported from 2026-07-09 (Wuxi); no named T1 customers public y

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
CASBOT W1 / W2 灵贰 CASBOT (Beijing Zhongke Huiling Robot Technology)	shipping	bipedal legs (NOTE: discovery misclassified 02 as wheeled; t	n/a	from RMB 328,800 on JD.com (02); 02 Lite dev platform RMB 99,000 promo	first JD orders within 48h of June 2025 listing; family robots working on Lenovo notebook assembly line; CASBO
U05 (UU U05) 优友 U05 Canbot	shipping	three-wheel wheeled base, auto-docking charging	n/a	HKD 320,000 (HK reseller, historical/out-of-stock)	hundreds of commercial deployments 2016-19 (banks, hospitals, airports, malls); National Day 70th anniversary,
W1 Pro 跨维W1 Pro DexForce	shipping	wheeled; omnidirectional (40 DoF config) or differential (34	10	n/a	vendor-listed partners: Huawei, BYD, Foxconn, Panasonic, Toyota, SOEs (aerospace/chemical/shipbuilding); coffe
Mercury X1 Mercury X1 水星X1 Elephant Robotics	shipping	wheeled AGV base (torso = Mercury B1)	1	USD 15,999 (X1); Mercury B1 torso ~EUR 17,599 at EU reseller	commercially listed at multiple international resellers since 2024; education/research users; volumes not disc
R1 Pro 星海图 R1 Pro Galaxea AI	shipping	wheeled omnidirectional; 3 self-developed steering-wheel mod	3.5	USD 69,999 (US reseller Robotopian)	research labs in CN + international; reference platform for Galaxea G0 VLA work; no industrial production depl
G1 Galbot G1 Galbot (Galaxy General)	shipping	wheeled omnidirectional base + single telescoping/folding le	5	JD.com RMB 690-700k (~USD 97k); flash sale RMB 630k 02/2026; US resell	~170 unmanned kiosk stores/pharmacies in 40+ cities; Haidian pharmacy license 2026-03-12; CATL/Mercedes-Benz C
S1 Galbot S1 (银河通用) Galbot / Galaxy General	shipping	four-wheel omnidirectional AMR base with mast	30	~USD 55,000 listed (inquire; likely marketing anchor)	reported in regular use at CATL battery factory production lines (Jan 2026 onward)
Juliet 朱丽叶 Haohai Xingkong (HHS Robotics)	shipping	wheeled composite humanoid (concealed wheeled base)	10	n/a	Nanjing Road cabin (2 units, from 2026-02-13): 61,198 visitors, 2,368 orders, 3,082 items sold during Spring F
e3 (e³-Mobile / e³-Desktop / e³-Mobile-π) e³ HexFellow Robotics	shipping	modular: TRIGGER-A3 3-omni-wheel holonomic (π variant) / MAV	2	not disclosed (quote-based storefront)	no named customers/unit counts; active GitHub (75 repos) and community
K1W 节卡 K1W JAKA Robotics	shipping	wheeled omnidirectional base + lifting/pitching waist	n/a	n/a	working in a Fortune-500 enterprise factory (sorting/handling/tending/inspection); 300+ embodied-family units
Kargo JAKA Kargo (节卡) JAKA Robotics	shipping	omnidirectional wheeled AMR base with integrated cargo deck	5	~USD 65,000	CIIF 2025 launch; no named Kargo customers yet (company: 300+ embodied units 2024, >10k cobots installed)
XMAN-R1 XMAN-R1 (擎朗具身服务机器人) Keenon Robotics	shipping	wheeled (2-DoF chassis, two wheel modules under kneeling-leg	3	~USD 100,000	Shangri-La Traders Hotel Shanghai Hongqiao commercial deployment (announced 2025-10-31) in multi-robot model w

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
VersaBot VB1 / VB1-I / VB2 VersaBot (蓝芯科技) Lanxin Robotics	shipping	VB1/VB2: four-wheel omnidirectional AMR; VB1-I: dual-wheel d	2	~USD 120,000 (third-party listing, unconfirmed)	Lanxin AMR/vision business deployed at Huawei, BYD, LG (vendor); no VersaBot-specific counts disclosed
MIRO U 美罗U Midea Group	shipping	wheeled(-legged) column chassis, 360° in-place rotation, six	n/a	n/a	Wuxi high-end washing-machine factory lines from end-2025 (own factory, self-reported 30% changeover gain); MI
Hobbs / Hobbs W1 Hobbs W1 (松延动力) Noetix Robotics	shipping	W1: wheeled omnidirectional platform (enclosed skirt base);	n/a	Hobbs bust ~USD 35,000; Hobbs W1 USD 73,500	500+ JD.com pre-sale bookings in first days (Dec 2025); pre-launch deployments in museums, government halls, o
Adam-U Adam-U PNDbotics	shipping	fixed pedestal (lifting column + chassis, ~40 kg counterweig	n/a	USD 45,000 (reseller in-stock)	research labs / data-collection use; public demos (dance, cloth handling); volumes n/a
TORA-ONE TORA-ONE PaXini Tech	shipping	wheeled base, laser-SLAM navigation	5	n/a	pilots claimed in logistics/industrial/commercial; no named customers; JD.com strategic investor; hands/sensor
D7 PUDU D7 (半人形) Pudu Robotics	shipping	wheeled (omnidirectional chassis)	n/a	not announced	no named customers or unit counts published; channel = 80k+ Pudu robots in 60+ countries
FlashBot Arm 闪电臂 Pudu Robotics	shipping	wheeled delivery-robot base (FlashBot platform), 539x515mm f	n/a	~USD 34,000 (unofficial; base FlashBot ~\$28k; official pricing unpubli	no named FlashBot Arm customers public as of 08/2026; base FlashBot widely deployed in hotels
Q5 星动Q5 Robot Era	shipping	wheeled, with 3-DoF flexing/height-adjusting leg section	10	n/a	mall guidance deployments; Chengdu Panda Base; Chengdu debut 2025-06-19 with 100+ intent orders; company-wide
X1-PRO (with X1 predecessor and C1-D universal humanoid chassis) 仙工 X1 PRO / C1-D 通用人形底盘 SEER Robotics	shipping	wheeled omnidirectional chassis with cargo buffer tray; sepa	6	n/a	wheeled humanoids delivered to customer sites for trials since late 2024; logistics pilots; WRC 2025/CeMAT 202
Moz1 小墨 (Xiaomo) Spirit AI	shipping	4-wheel mecanum omnidirectional wheeled base	n/a	~USD 150,000 (aggregator estimate)	CATL Zhongzhou battery-pack line: connector insertion for EOL/DCR testing, ~99% success, 3x efficiency (promot
Cruzr S2 克鲁泽S2 UBTech Robotics	shipping	wheeled (full-size wheeled humanoid; drive type not disclose	n/a	n/a	WRC 2025 multi-robot warehouse demo; Japan commercial sales launch Oct 2025; no named end customers/unit count
Wanda 2.0 万达 2.0 UniX AI	shipping	wheeled chassis (quiet, obstacle-avoiding; concealed drive w	n/a	from RMB 88,000 (pre-sale base config, 2024-12-25)	CCTV-broadcast home demos (laundry, burgers, tofu, dishwashing) 08/2024; WRC 2024 booth; pre-sales + mass prod

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
R1-D 宇树R1-D双臂机器人 Unitree Robotics	shipping	dual-arm upper body; fixed bench base (A5/A7) or wheeled bas	2-4 (max)	from RMB 26,900 / USD 4,290 (R1-A5)	launch-quarter product, no named customers yet; company shipped >5,500 humanoids in 2025 (32.4% global unit)
Tian Yi 2.0 天轶2.0 X-Humanoid (Beijing Humanoid Robot Innovation Center)	shipping	wheeled AMR-style base with articulated torso pitch linkage	n/a	~USD 60,000 (third-party listing)	Gold + silver medals, World Humanoid Robot Games Aug 2025 (27 autonomous 8mm precision insertions); JD.com ret
H1 Zerith H1 Zerith Robotics	shipping	universal-wheel compact chassis (corridor/elevator compatibl	n/a	n/a	20+ venues; tens of millions CNY orders, ~¥100M intention orders; 500-unit year-end target
CASIVIBOT CASIVIBOT (中科慧远) CasiVision	announced	wheeled AMR base (enclosed, drive type n/a)	n/a	~USD 29,000 (configuration-dependent)	launch media coverage only; no named customers; obvious channel = CasiVision's installed AOI base
Centaur-G1 Centaur-G1 Elite Robots	announced	mecanum-wheel omnidirectional base, ±10mm chassis positionin	5	n/a	none announced; company has ~10,000 cobots in 50+ countries as conversion base
AICO 2 爱可 AICO 2 Flexiv	announced	dual-wheel differential-drive AMR base, laser SLAM	4	n/a	no named AICO 2 deployments; MICO demoed ICRA 2026; company's Rizon arms widely deployed
GRW 傅利叶GRW 轮式双臂机器人 Fourier Intelligence	announced	wheeled (parallel mechanism, independent drive + steering pe	n/a	n/a	WAIC 2026 booth demos only; no customers/pilots disclosed yet
HIVA (Haiwa) 海娃 Haier (platform: Robot Era)	announced	wheeled (Robot Era Q5 platform)	n/a	n/a	AWE 2025 exhibition demos only
KNEWBOTS wheel-arm robot 晓悟智能轮臂式机器人 KNEWBOTS	announced	wheeled AMR chassis	4.5	n/a	no public deployments/unit counts; demo/pilot stage
RealBot-S2 / RealBot-L2 RealBot-S2 / RealBot-L2 RealMan Robotics	announced	S2: wheeled four-steer four-drive + folding leg; L2: wheeled	5	n/a	WAIC 2026 demos (fridge opening, cross-city teleop tea-making); company platforms in Yining Xinjiang restaurant
R1 蚂蚁灵波 R1 Robbyant (Ant Lingbo Technology, Ant Group)	announced	wheeled pedestal (single-column, circular base)	n/a	from ~USD 55,000	robot-chef demo (4 dishes) at Sep 2025 Bund Conference launch; reported pilots in Chinese museums, drugstores
Helios 轮式双臂机器人Helios Rokae Robotics	announced	omnidirectional wheeled base (3 DoF planar)	5	n/a	none disclosed; trade-show demos (Automate 2026 Chicago)
Onero H1 Onero H1 SwitchBot (OneRobotics)	announced	cylindrical wheeled base (vacuum-robot-style puck; drive lay	n/a	USD 9,999 official store listing ('coming soon'); CES press	CES 2026 live demos only (laundry folding, dishes); pre-production 'lab' status; no customer deploym

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
TWH020 (Xiaotuo) TWH020 小拓 Topstar Technology	announced	4 steering-wheel (四舵轮) omnidirectional chassis, 360°	10	n/a	claimed first humanoid in Chinese injection-molding scenarios; deployments inside Topstar automation projects;
SageMan (iSAGE STAR) 赛智四臂人形机器人 SageBot/SageMa iSAGEBOT / Shanghai SAGE Intelligent Technology	announced	wheeled (AMR-derived platform)	3-16 (configuration-dependent)	n/a	no confirmed paid humanoid deployments; 100+ LOIs claimed for six-arm MAHAKBOT at CTIF Shanghai
AstroD. AD-01 (AstroDroid) AstroDroid AD-01 (原力无限) Elu.AI / INFIFORCE	prototype	wheeled AMR base with articulated pitch linkage to humanoid	3	~USD 100,000 (inquire)	APSARA Conference demo (2025-09-24) only; no named customers
Futuring 2 未来不远 二代 Futuring Robot	prototype	wheeled base with slim column under dual-arm torso	n/a	USD 5,000 (inquire)	claimed 300+ Chinese households, 30,000+ service hours (vendor-claimed, echoed by Chinese press)
Maker H01 Maker H01 (极佳科技) GigaAI	prototype	omnidirectional all-wheel-drive chassis	5	~USD 160,000 (listed, unverified)	none disclosed; unveiled 11/2025
ProWhite 2.0 (+ Zeus 1) ProWhite Robot 2.0 (普罗宇宙) PL-Universe	prototype	four-wheel omnidirectional wheeled base	7	ProWhite: not disclosed (humanoid.guide \$45k entry conflicts with vend	none found; no named customers or pilots
Psi V1 Psi V1 (灵初智能) PsiBot / Lingchu Intelligence	prototype	wheeled (rounded chassis, hub-motor style side wheels in ren	n/a	n/a	small-scale warehouse sorting validation (vendor); mahjong/block demos; Meituan last-mile robot co-dev (separa
Vinci Vinci 轮式人形机器人 SVT Robot (Hangzhou Spacetime Variable Technology)	prototype	4-wheel-drive / 4-wheel-steer omnidirectional wheeled base (	n/a	n/a	none public; office teleop demo videos only (late 2025)
Songyi 松羿 Siasun Robot & Automation	prototype	wheeled (Siasun AMR-heritage mobile platform; drive type not	n/a	n/a	WRC 2025 exhibition demos only; no customers disclosed
I-Series (wheeled variant) Simplexity Robotics	prototype	omnidirectional mecanum-wheel base with articulated waist li	n/a	USD 78,000	none; traction is financial (~\$289M in 5 rounds, unicorn, Tencent/Alibaba/HongShan)
Quanta X2 量子X2 X Square Robot	prototype	wheeled (low sculpted chassis, concealed wheels)	6	~USD 80,000	demo videos only; no named customers found
Sean Sean (ZeroTh) Zeroth Robotics	prototype	wheeled pedestal (rounded enclosed base, corner wheels)	n/a	~USD 65,000	none; absent from vendor's current product line (M1/W1/Jupiter) - likely internal/dev platform

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
Zoomlion wheeled humanoid (unnamed, 2 models) 中联重科轮式人形机器人 Zoomlion	prototype	wheeled chassis with dual-arm humanoid torso (one variant ap	n/a	n/a (internal use, not for sale)	dozens of humanoids working in Zoomlion Changsha Smart City factories (machining, logistics, pre-assembly, QC)
ByteMini ByteMini ByteDance Seed	research	wheeled mobile base with lift column	n/a	n/a (internal research platform)	research demos only (table bussing, cloth manipulation, pick-and-place)
Ginger (XR-1 / Cloud Ginger 1.0 / 2.0) Cloud Ginger 云端机器人小姜 CloudMinds / Dataa Robotics	discontinued	3-wheel omnidirectional self-balancing chassis (skirted)	n/a	CNY 698,000 (JD.com listing)	COVID field hospitals (Wuhan/Shanghai), telecom stores, schools, exhibitions 2019-2023; volumes undisclosed
Sanbot Max (King Kong) Sanbot King Kong Qihan Technology	discontinued	four-wheel mecanum omnidirectional	n/a	n/a	Sanbot line deployed at Shenzhen Bao'an airport and Gongbei Port of Entry (mainly Elf); international dis

7 Regional landscape — USA & Canada

46 documented robots (incl. Canada) — the economics wave. The 2025–26 US generation sells labor, not robots: Sunday's Memo (Coatue-backed, \$1.15B, glove-collected ~10M episodes, <\$10k target, beta by Thanksgiving 2026), Weave's \$7,999/\$449-mo Isaac, Robot.com's R-Noid and Eden's \$10/hour RaaS, Dyna's laundry robots at 99%+ 24-h success, Reflex's teleop-to-autonomy warehouse play (GXO; sub-\$50k target; Mexico factory), and Dexterity's 60 kg "superhumanoid" Mech (Kawasaki-built arms, Sanmina manufacturing). Walden Robotics crystallizes the moment: Russ Tedrake's TRI spin-out emerged July 2026 at a \$1.1B valuation with robots already in production shifts at a Toyota plant. Public-market reality check: Richtech (NASDAQ: RR) pairs a \$180k ADAM install base with \$5M revenue and governance flags. Canada contributes Sanctuary's hydraulic-hand Phoenix (now software-pivoted) and Kinova's discontinued MOVO. The deep research heritage — PR2, Baxter, Mobile ALOHA (\$32k BOM that spawned Sunday and the ALOHA-clone industry), Robonaut 2, TRI Punyo — remains the field's intellectual substrate. The FCC Covered-List action (§14) now shelters this market from foreign hardware — and pressures everyone else's US strategy.

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
Aeo aeo Aeolus Robotics, Inc.	shipping	wheeled (drive type not disclosed; appears diff-drive with c	3.6	n/a (not disclosed); RaaS leasing/subscription only	since 2019: Medical Care Service Inc., Gakken Cocofump, HIMEDIC (eldercare); Tokyu Group, Globeship (property
Axol Almond (Almond Robotics)	shipping	standalone / Axol Mount (fixed) / Axol Cart (wheeled)	6.5	from USD 7,999; USD 11,999 full kit (cameras + Orin NX)	no named customers; batch sales to physical-AI teams; Product Hunt/press launch June 2026
Alex Boardwalk Robotics	shipping	pedestal (fixed) or third-party AMR mounting; no built-in lo	n/a	n/a	research-institution sales; aviation MRO pilot; carbon-fiber sanding demo
Simba (V1, Medium Payload Unit) Deft Robotics	shipping	wheeled (holonomic) AMR base with lift column	4	USD 34,900 (store 2026); earlier USD 40,000 + 30,000/yr maintenance; r	49 units sold (store counter); 2 paid enterprise contracts; Tier-1 US auto supplier case studies (compressor h
Vega Vega Dexmate, Inc.	shipping	wheeled (omni-directional)	4.5	USD 89,999 MSRP (\$999 preorder deposit); some third-party listings cit	in testing with unnamed publicly traded companies (manufacturing, logistics, retail); 6 Vega-U grant units to
Mech Mech Dexterity, Inc.	shipping	omnidirectional rover, 4 independently steerable wheels; spi	29	n/a	truck-loading app launched 03/2025; Sanmina scaling deal for parcel/e-commerce/retail/air-cargo customers; uni
DYNA-1 Dyna Robotics	shipping	stationary workcell (table/cart mounted); mobile base in dev	n/a	n/a (not disclosed); RaaS monthly 'sell labor not hardware'	hotels, commercial laundries, restaurants, gyms; 16+ h/day operation; 24h run 850-900+ napkins @ 99.4% success
R2D3 Open Droids	shipping	wheeled AMR (Woosh chassis, diff-drive est.)	5	USD 55,000-60,000	rehab-center pilots (vendor-claimed); Baskin-Robbins/Subway pre-order talks; no named paying customers

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
XLeRobot Open-source project (Vector Wang / Rice University RobotPi Lab)	shipping	wheeled (3 omni wheels, holonomic; IKEA RASKOG cart chassis)	n/a	from ~\$660 self-build; Wowrobo kit \$579 (CNY 3699)	~5.4k GitHub stars, ~590 forks (2026-08); community builds, household-task demo videos
Reflex Reflex Reflex Robotics	shipping	compact wheeled base, footprint <2x2 ft (<61x61 cm)	n/a	lease/RaaS only; reported target <USD 50,000/unit	GXO RaaS pilot since 2024-09-18 at Fortune 100 retailer omni-channel fulfillment site (tote transfers, picking)
ADAM ADAM Richtech Robotics Inc.	shipping	stationary pedestal / counter-integrated workstation (no mob	n/a	USD 180,000 list; RaaS USD 3,500/robot/month (25-unit/60-mo/\$5.25M dea	One Kitchen/Walmart ghost kitchens (Dawsonville GA 06/2024, Rockford IL 08/2024; 240-unit LOI; 20-restaurant f
TITAN RoboForce	shipping	interchangeable wheeled OR tracked mobile base (outdoor-rate	40	n/a	Intersolar Feb 2025 debut; 2025 customer field pilots; Three Rivers Solar demonstration; 11,000+ pre-orders vi
R-noid Robot.com (ex-Kiwibot)	shipping	holonomic wheeled base, 0.55 m footprint	4	RaaS subscription, rates not disclosed	<40 units across ~12 customers (launch, 2026); Harbor Links Golf Course NY disclosed; kitchens, laundries,
Mobile AI Trossen AI Mobile Kit Trossen Robotics	shipping	wheeled (differential drive, SLATE UGV, 2x 350 W hub motors)	1.5	USD 33,695.95 list / 22,995 sale (no laptop); 37,845.95 list / 27,145	successor to Stanford Mobile ALOHA; ALOHA-class kits widely used across academic labs, Google DeepMind and the
Operator OP1 Ultra (Ultra Robotics)	shipping	STATIONARY workcell on locking casters (no locomotion) - bor	n/a	RaaS USD 2,500-3,000/month; 3-month trial	Highline Commerce (Brooklyn 3PL): ~30% of linear fulfillment workflow automated in live operation; 'wareh
Walden W0.5 Walden Robotics	shipping	wheeled (legless; drive type not disclosed)	n/a	n/a	In production at a North American Toyota plant since 02/2026, 8-hour shifts; pilot-to-production <2 months;
Isaac 1 Isaac 1 Weave Robotics	shipping	wheeled base (3 DoF) with central column, passively stable	n/a	USD 7,999 up-front or USD 449/month; USD 250 refundable deposit	no Isaac 1 field units yet; predecessor Isaac 0 (stationary folder) shipping since 02/2026: 2,000+ field hours
Dex Dex Richtech Robotics Inc.	announced	wheeled AMR base (Richtech Titan delivery-series technology)	n/a	n/a	no customer announced as of 12/2025; demos: GTC 2025 DC baseball-cap assembly (booth 368), CES 2026
ROBIN Roboligent Inc.	announced	wheeled AMR base	n/a	USD 85,000 (aggregator-listed, unverified)	no named customers; AFWERX STTR I + \$1.8M II fund Optimo rehab arm; RobotWorld 2025 Tesollo demo
Memo Memo Sunday Robotics (Sunday Inc.)	announced	wheeled platform with central telescoping column (no legs)	n/a	not announced; ~\$20k hand-built cost today; >=50% cost cut at scale	no customer deployments yet; WIRED-observed live demos (espresso, dishwasher, glasses); 2,000+ paid glove data

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
Syntro (V17) WorkFar Inc.	announced	3-wheel omnidirectional skirted pedestal base	n/a	USD 9,999.99-59,999.99 by tier; USD 29,999 commonly cited; robot+opera	none independently verified; all claims from company PR; aggregator flags 'not verified'
Beomni 1.0 Beyond Imagination, Inc.	prototype	humanoid torso on wheeled mobile base or fixed workstation	n/a	n/a	CES 2022 + Explorers Club teleop demos; 2022 SELF Labs LOI >=1,000 robots (no visible follow-through); no i
Eno Genesis AI	prototype	wheeled base with foldable articulated tower torso	n/a	~USD 90,000 (aggregator-reported, unconfirmed by vendor)	LG CNS multi-phase pilot/scale partnership (LG US operations) announced at launch; no field deployments yet
KR1 / Kinisi 01 Kinisi Robotics	prototype	wheeled (omnidirectional, zero-turn round base)	n/a	USD 75,000	no named customers; Automate Start-up Challenge winner May 2025; ~\$2M funding; acquired by Bear Robotics 2026
Eggie Eggie Tangible Robots	prototype	wheeled (squat four-wheel base under column torso)	n/a	~USD 32,000	demo videos only (laundry folding, spill cleaning, plant watering); no customers or pilots
Tobi TwoLabs	prototype	wheeled box base with articulated spine/waist column	n/a	USD 50,000 (target)	none; no pilots or customers found; 2-person YC Spring 2026 team
Mobile ALOHA Stanford University (IRIS Lab)	research	wheeled (diff-drive, AgileX Tracer AGV)	1.5	~USD 32,000 self-build BOM; Trossen Mobile AI kit ~USD 23-34k	research only; replicated by many labs; DeepMind ALOHA 2 / Unleashed follow-ups; lineage to Sunday Robotics Me
Punyo Toyota Research Institute	research	stationary upper-body rig (no mobile base published)	n/a	n/a	lab demos (jugs, boxes, laundry baskets, handovers); peer-reviewed papers; Soft-Bubble design replicated by ot
Astra (with Apollo wheeled/pedestal variants) Appronik	discontinued	Astra: fixed stand or third-party AMR. Apollo: modular lower	10	n/a	Mercedes-Benz factory pilots since Mar 2024; Jabil plant pilot + contract manufacturing; GXO warehouse deploym
HERB 2.0 CMU Personal Robotics Lab	discontinued	wheeled (Segway RMP 200, 2-wheel + casters), 655x620mm footp	n/a	n/a	single research platform; decade of manipulation-planning publications; Intel + NSF QoLT funded
Cody Georgia Tech Healthcare Robotics Lab	discontinued	wheeled omnidirectional (Segway RMP 50 Omni, mecanum), 750x6	n/a	n/a	lab + patient studies only: bed-bath trials, robot-touch perception studies (2010-11)
RoNA (Robotic Nursing Assistant) Hstar Technologies	discontinued	wheeled omnidirectional (mecanum)	68	n/a	prototype demos and healthcare case studies only; TATRC/SBIR funded; no clinical deployment found
Domo MIT CSAIL Humanoid Robotics Group	discontinued	stationary upper torso (no mobility)	n/a	n/a	lab research only; 2007 PhD dissertation; lineage to Meka Robotics and Hello Robot

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
M1 Mobile Manipulator Meka Robotics	discontinued	wheeled holonomic omnidirectional (Meka B1 base), 460x670mm	n/a	USD 320,000-340,000 depending on configuration	few units to research labs; Meka modules widely used (GT Cody, UT Austin Dreamer)
Robonaut 2 (R2) NASA JSC + General Motors	discontinued	modular: ISS pedestal/stanchion; Centaur 2 4-wheel rover (Ea)	9	n/a	ISS 2011-2018 (task boards, switches; largely disabled 2015-18 by power fault); Centaur 2 desert trials 2010;
Guardian XT Palladyne AI (formerly Sarcos Technology and Robotics)	discontinued	platform-agnostic torso: mounts on tracked bases (Guardian X	n/a	never commercialized; RaaS subscription planned (XO was ~USD 100k/yr)	beta complete 2021, CES demos, Navy/USAF development work (incl. RE2 Sapien/HDMS); zero commercial deployments
Baxter Rethink Robotics	discontinued	stationary torso on optional caster pedestal (no powered mob	2.2	USD 22,000 base (2012); ~USD 25k with pedestal/options	thousands of Baxter+Sawyer shipped (vendor-claimed); SME packaging users; hundreds of university labs as defau
uBot series (uBot-5/6/7) UMass Amherst Laboratory for Perceptual Robotics	discontinued	two-wheel dynamically self-balancing (postural transitions t	n/a	n/a	lab research only: whole-body control, box flipping, postural transitions, first-responder teleop studies; bas
BEAR (Battlefield Extraction-Assist Robot) Vecna Technologies	discontinued	tracked, articulated dual tracked 'legs' with dyna	n/a	n/a	prototype demos: 227kg lift, dummy extraction, rubble traversal; soldier evaluations at Army Infantry Center ~
PR2 Willow Garage	discontinued	wheeled omnidirectional (4 powered caster modules), 668x668m	1.8	USD 400,000 (dual-arm); USD 285,000 single-arm SE; \$120k open-source d	11 free beta units (2010) + sales; ~50 units total at research institutions worldwide; no commercial use

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
MH3 Mirsee Robotics	announced	wheeled AMR base (skirted, bumpered); wheels chosen over leg	30	USD 200,000	none found; no customers, pilots or unit counts
Phoenix (Generation 8) Sanctuary AI	prototype	wheeled (small mobile base; legs dropped as 'too frail	n/a	n/a (not for open sale)	Gen 5 retail pilot Jan 2023; Magna plant trials 2024-2025 (material handling, sub-assembly, <24h task learn
MOVO Kinova	discontinued	wheeled holonomic (4 Swedish wheels), 508x762mm footprint, A	2.2	n/a	university beta community: UWaterloo RoboHub, Georgia Tech CORE, UW-Madison, EU labs; est. dozens of units

8 Regional landscape — Europe

38 documented robots — research superpower, fragmented commercially. Active products span PAL Robotics' TIAGo Pro/++/ARI (the EU research standard), Pollen/Hugging Face's open-source Reachy 2 (~\$70k), igus's €47,999 plastics-based Iggy Rob, Oversonic's RoBee (STMicroelectronics deal, US launch — but certifications are CE-machinery grade, not robot-safety), Enchanted Tools' ball-balancing Mirokai in French hospitals, and Duatic's ETH-lineage Alpha (12 kg continuous dual-arm payload, IP66). Humanoid (UK) became Europe's newest unicorn (\$1.35B; Bosch and Schaeffler backing; wheeled Beta units Q4 2026); Franka's Mobile FR3 Duo gives European labs a domestic reference platform; Wandercraft's Calvin (bipedal, hence annexed) took Renault's money into real plants; and Munich's RobCo — fresh off a \$100M Series C — announced its bimanual, humanoid-form "Alfie" in April 2026 (officially shown only from the waist up — though its promo video briefly reveals a mobile base in simulation), moving Germany's best-funded modular-robot startup into the category via RaaS. Germany specifically: Neura's MiPA is the declared household/care entrant (€9,999, deliveries slipping; Series C up to \$1.4B confirmed June 2026), Devanthro sells care-avatar telepresence, and the research bench — DLR's Justin, KIT's ARMAR-III, TUM's GARMI (no spin-off exists — available IP/talent), FZI's HoLLiE — is unmatched but under-commercialized. Cautionary tales are equally instructive: Aldebaran (Pepper) liquidated in 2025 with its assets sold to a Chinese buyer; Care-O-bot's maker became an AMR vendor; Cybedroid sold 6 units and died. A telling blank on the origin map: the Nordics. Norway created the archetypal wheeled humanoid (Halodi/1X's EVE) but the company relocated to Palo Alto and pivoted to legs; Sweden's entry (Hexagon's AEON, Schaeffler-backed) is a wheel-leg hybrid sitting in this study's borderline annex; and Denmark's Odense cluster — Universal Robots' arms, MiR's bases — supplies the category's components without ever assembling a semi-humanoid of its own. Europe's gap is not capability but industrialization — precisely the entry a well-capitalized German robotics company can make ahead of the 2027 Machinery Regulation.

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
Mirokai Mirokaï (characters: Mir Enchanted Tools)	shipping	ballbot (balances on a sphere driven by 3 omnidirectional wh)	n/a	~EUR 50,000 purchase + several hundred EUR/month subscription (2026);	ISIR Sorbonne (first delivery Jul 2024); AP-HP incl. Broca & Annie Girardot pilots; Institut du Cancer de
Ameca Ameca Engineered Arts	shipping	fixed pedestal (non-walking; Gen 2.6 articulated non-walking)	n/a	~USD 130k-250k full body (quote-only); head configs from ~\$25k; custom	National Robotarium Edinburgh, Computer History Museum, GSK trade shows, Sphere Las Vegas (EA robots); company
QTrobot QTrobot LuxAI S.A.	shipping	stationary desk pedestal (no locomotion)	n/a	Home: \$1,977 + \$149/mo support (2021); professional/research ~EUR 6,80	therapy centers/schools/universities across EU + US + Asia (trials in LU/FR/BE/DE from 2018; Univ. Birmingham,
Kime Kime Macco Robotics	shipping	fixed kiosk pedestal (~2 m2 enclosure); manually-moved cart	1	n/a	Seville bars (COVID contact-free beer service, 2020-21); Prio Portugal gas-station trial + planned rotation; S
InMoov Open-source project (Gael Langevin)	shipping	none official (torso platform); community DIY wheeled bases	n/a	under USD 1,000 in parts (2013 estimate) + printing	thousands of community builds worldwide since 2012 (builder map, schools, universities, exhibitions)
RoBee RoBee Oversonic Robotics	shipping	wheeled (diff-drive; omnidirectional on request)	5	n/a	STMicroelectronics global supply deal (first unit in ST Malta fab, Dec 2025); several Italian factories; hospi

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
ARI ARI PAL Robotics	shipping	wheeled (diff-drive, 2 DoF), skirt-covered base	n/a	not disclosed (quote-based)	SPRING H2020: 7 units, Broca gerontology hospital Paris (AP-HP), 100+ patients, completed Oct 2024; Ayesa rece
TIAGo++ / TIAGo OMNI++ TIAGo++ / TIAGo OMNI++ PAL Robotics	shipping	wheeled; differential drive (TIAGo++) or omnidirectional 4x	3	TIAGo++ from ~EUR 90,000 (2019, config-dependent); current pricing quo	40 collaborative projects, 14 EU countries, 27 countries worldwide (vendor); RoboCup@Home 2019 winner platform
TIAGo Pro TIAGo Pro PAL Robotics	shipping	wheeled omnidirectional (4x mecanum wheels, 4 DoF)	3	not disclosed (quote-based); ~USD 85-95k reported in market chatter, u	CO-HAND EU project: TRL 7 validation in textile manufacturing (2026); ROSALYA food-industry project; euROBIN r
Reachy 2 Reachy 2 Pollen Robotics (Hugging Face)	shipping	wheeled omnidirectional (3 omniwheels); stationary mount opt	3	~USD 70,000 (dual-arm + mobile base)	Cornell University, Carnegie Mellon University research labs; internal Hugging Face/LeRobot use; unit count un
Steve Sulu.be (Jan De Coster studio)	shipping	wheeled cart base (large front drive wheels + casters); huma	n/a	USD 60,000	character host in Belgian movie theaters (Brightfish), corporate receptions, festivals, brand activations; uni
iggyRob iggyRob (Iggy Rob) igus GmbH	shipping	wheeled AMR (ReBeLMove Pro, differential drive, three-point	2	EUR 47,999 (~USD 54,500); iggyRob Home variant price on request	no named external customers; igus internal use for injection-molding machine tending planned; live catalogue a
MiPA MiPA (My intelligent Per NEURA Robotics GmbH	announced	wheeled (actuated wheels, geometry not disclosed) + linear-a	3	EUR 9,999 introductory (MiPA Home, private households only); EUR 100 r	none verified; trade-fair demos only (Automatica 2025 beverage service, IFA 2025 tidying, CES 2026 dishwasher)
Alfie RobCo	announced	mobile base shown in simulation (promo video 0:14): broad fl	n/a	RaaS, terms undisclosed	none yet (Hannover Messe preview)
pib.Pro pib (printable intellige isento GmbH / isento robotics (pib.rocks)	announced	two-wheeled self-balancing (hoverboard-style) mobile base un	n/a	USD 5,999	DIY pib in German schools via pib.Education; no named pib.Pro customers yet
Clone Torso Clone Torso Clone Robotics	prototype	none — stationary mounted upper body/torso (full android Pro	n/a	not published for Torso; manufacturing cost <\$20k/unit (vendor clai	none beyond vendor videos (viral Torso/Protoclone clips); no independent evaluations or customer deployments f
Robody Robody Devanthro GmbH (The Robody Company)	prototype	wheeled mobile base (drive type not disclosed)	n/a	n/a — care-service model (Robody Cares), pricing undisclosed	Private-home deployments since early 2024 (3+ hw generations); Teleskoop BMBF project with FZI/Charité: househ
Alpha Alpha Duatic AG	prototype	wheeled (mecanum omnidirectional)	6	n/a (not disclosed); USD 150k third-party estimate	20 kg bag autonomous lift demo video; DynaArm demos on ANYmal and Festo BionicMobileAssistant; no named custom
Eden I Eden Robotics, Inc.	prototype	floor-standing rounded base + lift column (seen in site vide	n/a	RaaS from USD 10/hour, USD 0 setup; no purchase option	none public as of Aug 2026

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
Mobile FR3 Duo Franka Robotics	prototype	wheeled (omnidirectional TMR base, 800x580x294 mm)	3	not published (early access)	Automatica 2025 showcase; no commercial deployments disclosed
HMND 01 (wheeled) / Alpha Wheeled HMND 01 Alpha Wheeled Humanoid (Humanoid Ltd / SKL Robotics Ltd)	prototype	wheeled (omnidirectional pedestal base)	n/a	n/a (not disclosed); RaaS subscription incl. fleet software, maintenanc	PoCs: Ford Cologne (97% pick reliability, 83 picks/h, third-party); SAP+Martur Fompak (autonomous, SAP EWM-dri
Alfie Alfie Prosper Robotics	prototype	wheeled (indoor, single-floor)	n/a	target ~USD 5,000-15,000 hardware + ~USD 500/month subscription (maint	prototypes tested in Prosper employees' homes; no customers or unit counts; launch slipped from ~2025
ALPHA (Automated Litter Processing Humanoid Assistant) TeknTrash	prototype	wheeled AMR base with tall lift column and integrated collec	n/a	USD 55,000	pilot deployments with waste- management operators claimed; no named sites or counts
Northstar Northstar UMA	prototype	wheeled (V0 initial platform; bipedal only on later roadmap	n/a	~USD 70,000 (discovery sweep, unconfirmed)	none yet; ~50 potential customers evaluating (vendor-claimed); tethered prototype at July 2026 unveiling
Rollin' Justin Rollin' Justin (Agi DLR Institute of Robotics and Mechatronics	research	wheeled, 4 independently steered spring- suspended wheels on	14	n/a (not for sale)	METERON SUPVIS-Justin ISS teleop 2017 (Nespoli); Surface Avatar ISS campaign 2023-2025 (Rubio, Dyson/Epps, Kim
HoLLiE (HoLLiE C) HoLLiE — House of Living FZI Research Center for Information Technology	research	Clearpath Ridgeback omnidirectional (mecanum) commercial pla	n/a	n/a (not for sale)	HoLLiECares (BMBF): two field-test phases in Städtisches Klinikum Karlsruhe and Knappschaftsklinikum Saar (~20
R1 R1 — your personal human Istituto Italiano di Tecnologia (IIT)	research	two driving wheels (diff-drive) + 3 pivoting 75 mm castors	1	target 'cost of a small family car' (~EUR 20-30k class, esti	rehabilitation-assistant trials with Fondazione Don Gnocchi (assistive- rehab joint lab); HR1 healthcare assist
ARMAR-7 ARMAR-7 KIT H2T (High Performance Humanoid Technologies Lab)	research	holonomic wheeled platform, 3 omni- wheels	n/a	n/a (not for sale)	Automatica 2025 + Hannover Messe 2026 live demos (dishwasher unloading, laundry sorting, learning from demonst
GARMi GARMi TUM MIRMI (Geriatrics)	research	wheeled mobile base (stable, low CoG) + extendable lifting c	n/a	n/a (not for sale)	model apartment trials at Geriatrics centre Garmisch- Partenkirchen with Caritas/LongLeif; telemedicine and r

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
Alter-Ego (Alter-Ego X) ALTER-EGO Univ. Pisa Centro Piaggio + IIT SoftBots	research	two-wheel self-balancing (inverted pendulum)	n/a	n/a (not for sale)	pilot clinical avatar study at IRCCS Maugeri Milan (Fit4MedRob/PNRR): patient welcome, pain assessment, accomp
EVE EVE 1X Technologies (ex-Halodi Robotics)	discontinued	two-wheel self-balancing wheeled base, multi-terrain wheels	n/a	RaaS: ~NOK 500,000 (~USD 50k)/unit/year under 2022 ADT agreement	Everon/ADT EvoGuard night guarding, 140 units/yr lease (2022), est. 150-250 units cumulative; pilots: Altopack
Leenby Leenby Cybedroid	discontinued	wheeled base under skirt (drive type not disclosed)	n/a	n/a	~10 units total by 2019: 6 sold (companies, nursing homes, research), 3 event rentals, 1 six-month Crédit Agri
AILA AILA (Artificial Intelli DFKI Robotics Innovation Center)	discontinued	six-wheeled omnidirectional base, each wheel a 2-DoF (drive+	n/a	n/a (never for sale)	research projects only: SemProM, Robofoot (2010-2013), BesMan (2012-2016, ISS mockup/SpaceBot Cup), HySociaTea
Care-O-bot 4 Care-O-bot 4 ("Paul Mojin Robotics (now 4am Robotics) / Fraunhofer IPA	discontinued	wheeled omnidirectional	n/a	USD 66,000-232,000 by configuration (historical)	'Paul' greeter at Saturn stores from Oct 2016 (Ingolstadt first); museum/airport/care trials; small
AMIGO AMIGO (Autonomous Mate f Tech United, TU Eindhoven	discontinued	four omni wheels (soccer-robot drive technology); base house	1.5	n/a (self-built)	RoboCup@Home: Dutch Open champion 2012, world runner-up 2014 (Brazil); RoboEarth + Bobbie Robotics demonstrato
Cosero Cosero (Cognitive Servic University of Bonn — NimbRo	discontinued	omnidirectional: 4 individually steerable wheel-pair axes, 8	n/a	n/a (self-built)	RoboCup@Home world champion 2011-2013 + German Opens 2011-2014; demos: omelet cooking, fridge access, table ca

9 Regional landscape — Japan

33 documented robots — the longest institutional memory, now re-mobilizing. Japan invented the category's ancestors (wakamaru 2005, enon, SmartPal, Twendy-One, ROBEAR) and its icon: Pepper (~27,000 produced — still the largest semi-humanoid fleet ever; B2B support continues in Japan, with a "Pepper+" rental relaunch in 2026 even as Aldebaran died in Europe). Today's activity clusters in: **teleoperation-first services** (ugo Pro in building security; Ory's OriHime-D café employing 70+ disabled pilots — a social-innovation model unique to Japan; Jinki Ittai's "humanoid heavy machinery" in revenue service on JR West railways); **industrial dual-arm incumbents** (Kawada NEXTAGE since 2009; Yaskawa's MOTOMAN NEXT AI line — and note Yaskawa wholly owns Tokyo Robotics, whose Torobo2 builds Waseda's AIREC care robot: one keiretsu quietly spans Japan's flagship care program and its industrial dual-arm line); **hospital/care pilots** (Kawasaki Nyokkey at Fujita Health University; Foxconn/Kawasaki Nurabot launching FY2026); and **Toyota's ELEY** — QDD actuators in every joint, a scapular axis, and a production-site learning agenda: the highest-credibility incumbent program worldwide (§12). Tracked outlier: Mitsubishi Electric's Diaroid. Japanese firms also increasingly distribute Chinese wheeled humanoids (Cruzr S2 via GA Robotics), importing the competition.

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
Zero-shiki Jinki ver.2.0 / Multifunctional Railway Heavy Machine 零式人機 ver.2.0 / 多機能鉄道重機 Jinki Ittai (developer/licensor) + Nippon Signal (manufacturer) + JR West (customer)	shipping	humanoid torso on telescoping crane boom of road-rail vehicle	n/a	n/a	revenue service on JR West operating lines since 07/2024 (painting, tree cutting); operated by Nishi-Nippon EI
NEXTAGE / NEXTAGE Fillie ネクステージ / ネクステージ フィリー Kawada Robotics	shipping	fixed pedestal (manual repositioning between cells; no mobil	2	Fillie from ¥6,000,000 (pedestal, hand cameras optional extra)	100+ companies since 2009 (third-party DB); electronics/auto/cosmetics/food; GLORY Saitama cell lines (est.);
OriHime-D オリヒメ D (分身ロボット) OryLab Inc. (Ory Laboratory)	shipping	wheeled base (forward/backward/rotate, teleoperated)	n/a	n/a (service/employment model, not sold retail)	Avatar Robot Cafe DAWN ver.B permanent in Nihonbashi since 2021-06-21; 70+ disabled pilots employed (02/2024);
Foodly フードリー RT Corporation	shipping	pedestal on casters (manual repositioning, no self-drive)	n/a	n/a (made-to-order, customized); monthly rental available	Hirai (Kumamoto) bento line; Ichibiki (Aichi) retort side-dish line; Will of Factory staffing partnership; MET
Sciurus17 シューラス17 RT Corporation	shipping	bench-mount torso; optional S17-LIFT 700mm lift column docki	0.5	JPY 3,300,000 base; OP1 head JPY 55,000; Lift and Kachaka base separat	standard lab platform at Japanese universities/companies since 2018; unit counts not disclosed
SEED-Noïd (SEED-R7 series) シードノイド (SEED-R7シリーズ) THK / SEED Solutions	shipping	modular: SEED-Mover omnidirectional cart (4 omni wheels) + S	n/a	n/a (quotation-based; units sold individually)	NTT Communications ExTorch program adoption (2021-05); WRS-era research platform; academic use; no public unit

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
Torobo (Torobo2) トロボ Tokyo Robotics Inc.	shipping	wheeled omnidirectional cart (4-axis)	7	n/a (quotation)	shipping to research labs; SusHi Tech Tokyo 2026 exhibit; no public unit counts
Robovie-R3 ロボビー・アール3 (Robovie-R Ver Vstone (ATR lineage))	shipping	wheeled differential drive (2x40W); optional omnidirectional	n/a	JPY ~3,800,000	widely used in Japanese HRI research (ATR field trials in malls, schools, museums); care-facility and retail e
ugo Pro ugo Pro (ユーゴー プロ) ugo, Inc. (ex-Mira Robotics)	shipping	wheeled (drive type n/a; diff-drive est.) + torso lifter	n/a	n/a (RaaS via partners, 'negotiable'); armless sibling ugo E	Teikyo Univ Itabashi (05/2024), Yanmar FLYING-Y Bldg Osaka (11/2024), Mainichi Newspaper Bldg Osaka (04/2025),
AGIRobots Worker AGIRobots Worker (車輪型セミヒ AGIRobots Inc.	announced	wheeled — swerve-type steered-wheel modules, omnidirectional	4	n/a (not disclosed)	none; basic development complete; Robot Technology Japan 2026 live demo planned
Modular O-ID	announced	wheeled base with wheel hub motors (flat-floor); humanoid up	n/a	USD 70,000	none found
MOTOMAN NEXT-NHC10DE MOTOMAN NEXT-NHC10DE Yaskawa Electric	announced	stationary dual-arm (pedestal/cell mount; no mobile base sho	10	n/a	iREX 2025 picking/packing demos only; no customers announced. Lineage traction: MOTOMAN-SDA 15-axis dual-arms
Nurabot Nurabot (護理機器人) Foxconn (Hon Hai) + Kawasaki Heavy Industries, AI by NVIDIA	prototype	wheeled AMR (Nyokkey-derived) + dual-arm torso + cargo compa	n/a	n/a	Taichung Veterans General Hospital field trials since 04/2025 (delivery, guidance, night patrol); claimed 20-3
Nyro Individual maker (Ryota Kobayashi)	prototype	none (upper body on camera tripod; no mobile base yet)	0.5	n/a	none; project page 436 views / 5 likes (niche hobbyist visibility)
Nyokkey ニョッキー Kawasaki Heavy Industries / Kawasaki Robotics	prototype	wheeled AMR base + humanoid torso	6	not for sale (pilot placements only)	Fujita Health Univ Okazaki Medical Center: specimen transport + auto-loading into clinical analyzers with FORR
DiaroiD ディアロイド Mitsubishi Electric	prototype	tracked — CuboRex CuGo MEGA electric crawlers (independent b	20	n/a (not commercialized)	no commercial deployments; verified tasks: plug insertion ~4min, M6 bolt fastening, zipper-bag opening ~2min;
Honda Avatar Robot Honda アバターロボット Honda Motor Co., Ltd.	research	wheeled mobile base under dual-arm upper body	n/a	n/a (research program)	none; press demos (pull-tab can, ball force control, tool use) 2022; FY2023 technology verification
ibuki イブキ Osaka University Ishiguro Lab / JST ERATO	research	wheeled mobility unit: 2 active wheels + eccentric wheels (h	n/a	one-off research platform, never sold	lab and public demonstrations (hand-holding walk demos, telepresence studies); no deployments
Panda dual-arm mobile manipulator PRIN Lab (NII)	research	omnidirectional (TriOrb ball-drive base)	3	n/a (self-integrated)	lab use; public demo video

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
ELEY エレイ (Embodied LEarning r Toyota)	research	wheeled (rounded mobile base per official render)	n/a	n/a (research)	WRS 2025 Aichi exhibit; production-site pilots planned; HSR-derived tech already fielded (Potaro hospital tran
AIREC (Dry-AIREC ver. 1.0) アイレック (AI-driven Robot f Waseda University Sugano Lab / JST Moonshot Goal 3	research	wheeled round mobile base with articulated single-column low	n/a	target >= JPY 10M (~USD 67k) at ~2030 introduction	care-facility trials since 2024; demos: patient rolling, supine-to- sit transfer, sock dressing, scrambled eggs
TWENDY-ONE トウ エンディワン Waseda University Sugano Laboratory	research	omnidirectional wheeled cart	n/a	n/a (never sold)	none commercial; landmark demos 2007-2015: bed-to- wheelchair transfer assist, breakfast preparation, handovers
enon エノン Fujitsu Frontech / Fujitsu Laboratories	discontinued	wheeled (two drive wheels, 2 DoF per wheel)	0.5	individual quotation; no list price	pilot deployments: shopping- center guidance, showrooms, Kyotaro Nishimura Museum guide (2007); unit count neve
G1 GITAI G1 GITAI	discontinued	humanoid torso, fixed/rail mount (microgravity- oriented, no	n/a	n/a (never sold)	JAXA Kibo mock-up test 2018-12: 72% (13/18) astronaut tasks completed; JAXA joint research 2019-03; no flight
MELTANT-beta MELTANT-β (メル タント・ベータ) MELTIN MMI	discontinued	pedestal/frame-mounted upper body (no self- locomotion)	n/a	never sold	field trials in construction/plant settings; JAXA ISS crew-task avatar tests; AVATAR X program (ANA/JAXA); G20
wakamaru ワカマル Mitsubishi Heavy Industries	discontinued	wheeled (two drive wheels + casters), 450 mm base diameter	0	JPY 1,575,000 incl. tax (2005 retail); rental JPY 20,000- 120,000/day (	100-unit household test sale (Tokyo, 2005); ~30 event deployments incl. Expo 2005 Aichi; corporate reception r
ROBEAR ロベア RIKEN / Sumitomo Riko	discontinued	wheeled base with extendable stabilizing outrigger legs	n/a	never sold	lab demonstrations only (patient lifting, standing assistance); no field deployment
WorkSense W-01 ワークセンス W-01 Seiko Epson	discontinued	passive wheeled stand (manually repositioned; no autonomous	n/a	quotation-based, never published	at least one global manufacturing customer piloting in Europe (2018); no install-base figures published
Pepper ペッパー SoftBank Robotics (orig. Aldebaran / SoftBank Robotics Europe)	discontinued	wheeled (3 omnidirectional wheels)	n/a	2015: ¥198,000 + 36- mo subscription ≈ ¥1.08M 3-yr; Biz ~\$440-550/mo; 2	~27,000 produced / ~17,000 sold, 70 countries; Hamazushi all branches, banks, Pepper PARLOR cafe, Prague &
Model-T モデルT Telexistence Inc.	discontinued	wheeled base + vertical axis under dual-arm torso	n/a	n/a (service model, never sold)	FamilyMart Toshima Ecomusee Town trial (summer 2020); Lawson Tokyo Port City Takeshiba (opened 2020-09-14); su
Robina ロビーナ (TPR-Robina) Toyota Motor Corporation (Partner Robot division)	discontinued	wheeled under skirt: 2 drive wheels + 4 maneuvering wheels;	n/a	never sold (2 units built)	continuous public tour-guide service at Toyota Kaikan Exhibition Hall 2007-2009 (+twin thereafter) — among the

10 Regional landscape — South Korea

16 documented robots — chaebol-scale bets plus an explicit 2026 wheeled wave. Rainbow Robotics' RB-Y1 (\$80k research/\$120k commercial) is the flagship: Samsung took 35% control for \$181.6M, consolidated it, and stood up a Future Robotics Office; Coupang logistics pilots are confirmed. ROBOTIS ships the fully open-source AI Worker (swerve drive, imitation-learning workflow); XYZ's DEUX (\$29,900 fixed / \$39,900 mobile, 3-finger hands + Glove X data pipeline) targets retail restocking; LG's CLOiD (CES 2026, "Zero Labor Home", pilot production July 2026) attacks the home — while LG hedges via investments in AgiBot and Dexmate. The March-2026 "wheeled semi-humanoid wave" (Seoul Economic Daily) added listed incumbents T-Robotics ("will never build legs") and I.L.'s ILBOT (mass production since Feb 2026), plus Neubility's announced Billy and Neuromeka's cart-based EIR. Hyundai's DAL-e and legacy research (KIST, KAIST M-Hubo, NAVER AMBIDEX) round out a market whose defining feature is vertical integration into memory-chip-grade manufacturing supply chains.

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
AI Worker (FFW-SG2 / FFW-BG2) AI워커 ROBOTIS	shipping	wheeled (swerve drive, 3 independent steer+drive modules) —	3.0	FFW-BG2 ~USD 40,000 incl. VAT (EU reseller); quotes via ROBOTIS Americ	Reuters May 2026: trained on skilled workers' motions for industrial tasks; K-Humanoid Alliance member; u
RB-Y1 이동형 양팔로봇 RB-Y1 Rainbow Robotics	shipping	wheeled (2-wheel drive; optional Mecanum omnidirectional kit	3	USD 80,000 (research) / USD 120,000 (commercial), VAT excl.	130+ cumulative units (2025); MIT/UC Berkeley/UW/Georgia Tech; Coupang fulfillment-center pilot CONFIRMED (fir
ALICE M1 앨리스 M1 A Robot / AeiROBOT	announced	wheeled AMR platform (drive layout not disclosed) + double p	n/a	n/a	HL Mando Wonju steering plant, 1 pilot unit Dec 2025 (MOTIE M.AX program)
CLOiD LG 클로이드 LG Electronics	announced	wheeled autonomous base, low center of gravity	n/a	n/a (not disclosed); USD 15,000-25,000 industry estimate	CES 2026 live demos (fridge retrieval, oven loading, laundry folding); pilot production Jul 2026; no customers
Billy 빌리 Neubility	announced	wheeled (4-wheel vehicle base with cargo compartment)	n/a	n/a	none yet (concept stage)
EIR 에이르 Neuromeka	announced	wheeled cart-based mobile base (explicitly no legs; installa	n/a	n/a	assembly verification on electric-motor production lines (cable insertion, precision assembly); field trials w
DEUX 듀스 (Deuce) XYZ Inc.	announced	wheeled (swerve drive, 360° omnidirectional; fixed-base vari	5.5	Mobile DEUX USD 39,900; fixed DEUX USD 29,900; Glove X USD 3,900 (pre-	Live demo at LoungeX Seongsu store 2026-07-27 (packing, tidying); no external customers yet; Series B KRW 13B
Friday Friday Holiday Robotics	prototype	wheeled mobile base (enclosed, compact two-section chassis)	n/a	~USD 200,000 listed; mass-production target ~KRW 100M (~\$72k)	no named customers; Series A ~KRW 155bn (Jul 2026); manipulation demo videos
ILBOT 아이엘봇 I.L	prototype	wheeled (legless mobile base)	n/a	n/a	2 units in mass production at group affiliate IL Mobility (automotive injection-molding part sorting), from Fe

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
TR Works TR웍스 T-Robotics	prototype	wheeled (company's own AMR mounted beneath humanoid tor	n/a	n/a	none yet; first public showing AW 2026
ALLEX 알렉스 WIRobotics	prototype	pedestal (upper-body system); wheeled mobile config and bipe	3	USD 180,000	none disclosed; RLWRLD physical-AI partnership
M-Hubo KAIST Hubo Lab	research	wheeled (omnidirectional base)	10	n/a	public exhibition demo: beverage fetching at 24% human speed, 90% success controlled / 80% dynamic
AMBIDEX 앰비덱스 NAVER LABS	research	fixed pedestal/stand (upper-body dual-arm research robot, no	3	n/a	cleans Rookie delivery-robot fleet in NAVER 1784 HQ; IROS 2018 challenge winner; no external installs
DAL-e 달이 Hyundai Motor Group (Robotics Lab)	discontinued	wheeled, 4 omnidirectional wheels	0	n/a	pilot at Hyundai showroom, southern Seoul, from 2021-01-25; no broad rollout evidenced
CIROS KIST	discontinued	wheeled (omnidirectional)	n/a	n/a	exhibition demos only (Robot World 2012 salad-making)

11 Regional landscape — Rest of world

22 documented robots. Israel: Unlimited Robotics' Gary (~10 units in a Philadelphia hospital plus Israeli MoH deployments) — and Mobileye's \$900M acquisition of Mentee (bipedal, out of scope) shows the exit ceiling. India: from Invento's wound-down Mitra to Genrobotics' 200-kg-payload tracked GenBot, Perceptyne's assembly torsos and ISRO's space-bound Vyommitra (Gaganyaan G-1, now Q3 2027). Russia: Promobot's V.4 service fleet (800+ units claimed) persists, but its Robo-C android line is discontinued and the FCC action closes the US. Turkey's AKINROBOTICS, Saudi Arabia's QSS, Nepal's Paaila and Vietnam's defunct TOPIO show the long tail. The open-source movement matters strategically: XLeRobot (~\$660, 5.4k GitHub stars), BamBot (~\$300), InMoov and hobbyist builds like Nyro (differential-gear joints, QDD motors) are training a generation on semi-humanoid architecture — tomorrow's engineers and tomorrow's expectations.

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
ADA-7 ADA-7 Sosyal Robot AKINROBOTICS (AKINSOFT)	shipping	wheeled (concealed base), 360-degree rotation on the spot	n/a	n/a (sale and rental via akinoid.com, prices on request)	Mini ADA sibling on duty at Istanbul Airport (info/wayfinding); ADA family at Istanbul Robot Museum; fairs/TV/
Abi Andromeda Robotics	shipping	wheeled	n/a	~USD 25,000	Australian aged-care pilots (vendor-reported)
Sophia Sophia (索菲亞) Hanson Robotics	shipping	wheeled tripod (passive standard; motorized RC or autonomous)	0.5	n/a (never officially published); third-party estimates USD 30k-100k+;	appearances in 65+ countries; UN/ITU AI for Good fixture incl. 'Sophia and Robert' (Jul 2026); R&
BamBot B0 Open-source project (Tim Qian)	shipping	wheeled (small differential base, 3D-printed)	n/a	~USD 300 in parts	~956 GitHub stars / 93 forks; community builds; viral demo 2025-03
Ginger (Nepal) Paaila Technology	shipping	wheeled (concealed base under humanoid torso)	n/a	n/a	3 Ginger + 2 Ferry robots serving at Naulo restaurant since 2018; Pari (2 units/generations) at Nepal SBI Bank
Promobot V.4 Промобот V.4 Promobot	shipping	wheeled (2-wheel diff-drive + support casters)	0	from USD 25,000; ~\$2,000/day rental	800+ units / 40+ countries (vendor claims 43 countries); Perm MFC civil-servant robot 30k+ inquiries; Kuwait O
Gary Gary Unlimited Robotics	shipping	wheeled (four-wheel skirted base)	n/a	~USD 30,000 + monthly fee (hospital model); 2021 consumer plan \$99 wai	~12 units at Rabin + Shamir medical centers since early 2024; Israel MoH supply agreement; ~10 units at a Phil
TARA GEN1 iHub Robotics (IHUB Research and Robotics)	shipping	wheeled circular base under skirt cowl	n/a	USD 13,998	exports to UAE/Saudi Arabia reported, no unit counts or named customers; listing 'Not Verified'
Spaceo M1 / Spaceo Pro Muks Robotics	announced	wheeled box base (M1); wheeled chassis with adjustable heigh	10	M1 from ~USD 11,000; Pro USD 60,000	none found; no named customers or pilots
PR-34D Perceptyne	announced	stationary pedestal/pillar mount (no mobility by design)	6	~USD 120,000	pilots with unnamed Indian automotive/electronics manufacturers; \$3M seed Oct 2024 (Endiya, Yali, Whiteboard)

Robot / Company	Status	Base	kg/arm	Price	Deployment evidence
Bimanual Svaya Robotics	announced	stationary pedestal (mountable on third-party mobile bases a	10	~USD 90,000	no named customers/units; company credibility via DRDO quadruped+exoskeleton demonstrators and SR-L cobot line
GenBot Genrobotics (Genrobotic Innovations)	prototype	tracked - rubberized quad-track chassis with adaptive multi-	n/a	n/a	none for GenBot; company's Bandicoot sewer robots deployed in 22 Indian states + 4 countries
Vyommitra व्योममित्र ISRO (Vikram Sarabhai Space Centre)	prototype	none - legless seated half-humanoid (head + torso + 2 arms),	n/a	n/a	flight integration for Gaganyaan-1 since 2026-04-28; G1 launch targeted Q3 2027 (risk 2028; slipped from H2 20
Sara سارة QSS AI & Robots	prototype	concealed legless lower body (estimated wheeled pedestal)	n/a	n/a	event appearances: LEAP, DeepFest, AI for Good, Arabs Got Talent; QSS plan for up to 10,000 humanoids in 5 yea
TM Xplore 1 Techman Robot	prototype	wheeled	n/a	~USD 25,000 (indicative)	none public
Nadine NTU Singapore (IMI); hardware by Kokoro (Japan); research continued at MIRALab Geneva	research	seated/stationary android (no locomotion)	n/a	n/a	ArtScience Museum 2017 (>100k interactions); AIA Singapore customer service; Bright Hill Evergreen Home eld
KIKI КИКИ Alfa Robotics	discontinued	wheeled (concealed skirt-style mobile base)	n/a	n/a	promo/event deployments in Russia claimed; no counts or named customers found
KP-BOT Asimov Robotics Pvt Ltd	discontinued	wheeled (concealed base under skirt-style lower body)	n/a	INR 21.24 lakh (~USD 25,500)	1 unit at Kerala Police HQ 2019--2021; withdrawn to Cyberdome for training/research by 2025
Mitra मित्र Invento Robotics	discontinued	wheeled (concealed base, layout not disclosed)	0	n/a (est. ~USD 10-15K class per period press)	GES 2017 Modi/Ivanka greeting; banks/hotels/malls/cinemas/airports 2017-21; COVID patient telepresence at Yath
Robo-C / Robo-C2 Робо-С / Робо-С2 Promobot	discontinued	stationary pedestal/seated android (no locomotion)	1	historically ~USD 20,000-50,000 depending on customization	small numbers since 2019: service halls, museums, expos, celebrity doubles; Perm State Medical University pati
TOPIO Dio TOSY Robotics JSC	discontinued	wheeled (3-wheel omnidirectional platform)	n/a	n/a	trade-show demos only (AUTOMATICA 2010); no customers documented

12 Technology trends

Foundation models made the category investable

Vision-language-action (VLA) models are the demand engine: Physical Intelligence's $\pi 0/\pi 0.5$ run long-horizon household tasks on *wheeled* mobile manipulators (\$5.6B valuation, ~\$1B raise in talks); NVIDIA's GR00T, Google's Gemini Robotics 2 (whole-body control, adapts to new hardware with <200 examples), AgiBot's GO-2, Galbot's GraspVLA (billion-frame synthetic pretraining) and Figure's Helix define the stack. Consequence: hardware is increasingly judged as a **data-collection and policy-execution vehicle**. AgiBot World's million-trajectory dataset was collected on wheeled G2s; Sunday's 2,000 paid "memory developers" wear gloves whose geometry its robot hand mirrors; China operates 40+ state-funded training centers; 1X's EVE security fleet trained NEO. Teleoperation is not a stopgap but a revenue-generating data business (Reflex, ugo, Beomni) — Interact Analysis expects deployments to remain "predominantly semi-autonomous" near-term.

Hardware convergence

The 2026 median semi-humanoid: two 7-DoF arms; 20–40 total DoF; 3–5 kg per-arm payload (10–20 kg for logistics classes; outliers: Dexterity 60 kg, Duatic 30 kg peak, Realman 5 kg official); 1.3–1.75 m height, often with a lift column or folding torso (floor-to-2 m+ work envelopes: Galbot 2.4 m, AI² 0–2.4 m, Dexmate 0.66–2.2 m); omnidirectional wheels at ~1.5–2 m/s; 5–8+ h runtime with hot-swap batteries; 3D lidar + multi-RGB-D + wrist F/T sensing; NVIDIA Jetson Orin (now Thor: Richtech Dex, Dexmate) as de-facto compute; ROS 2 plus a proprietary autonomy/VLA layer. Actuation splits between QDD/backdrivable philosophies (Toyota ELEY, Duatic DynaArm, Spirit AI full force control) and harmonic/planetary precision claims (± 0.03 –0.1 mm: JAKA, Elite, iSageBot).

Hands and end-effectors

Parallel grippers still dominate deployments; dexterous hands are differentiating fast: Chinese five-finger hands now start under \$1,400 (Inspire ubiquitous; the market grew ~150% YoY into a "mass-production race"), Western options run to Shadow's ~\$300k; in-house designs (Pudu DH11, XYZ's 7-DoF 3-finger + Glove X, Sanctuary's hydraulic 21-DoF, PaXini's ITPU tactile arrays) anchor product identity. Tool-changers and media at flange remain rare disclosures — an open differentiation axis for industrial-grade entrants. Notably, several "humanoid" vendors ship off-the-shelf EEs (1X EVE used a Robotiq 3-finger; OpenDroids integrates RealMan hands).

13 Pricing & business models

Band	Examples (2026 street/list prices)	Model
<\$5k	Unitree R1-D ¥26,900 (~\$4.3k); open kits (XLeRobot \$579–660, BamBot ~\$300)	Entry/education; ecosystem seeding
\$5–30k	Weave Isaac \$7,999 or \$449/mo; Almond Axol from \$7,999; Atribot T1 ~\$14k China / \$40,950 via Western resellers; Elephant Mercury X1 \$15,999; SwitchBot Onero H1 target ~\$1–1.5k (pre-production); WorkFar tiers \$10–60k; XYZ DEUX \$29.9–39.9k; Techman ~\$25k est.	Home subscriptions; research/edu; prosumer
\$30–100k	Deft Simba \$34,900; igus Iggy Rob €47,999; AgiBot A2 Lite \$44.6k; A2-W ¥400–800k insider; Atribot S1 ~\$69k; Reachy 2 ~\$70k; UMA Northstar ~\$70k; Noble \$75k; Rainbow RB-Y1 \$80k research; Dexmate Vega \$89,999; Genesis Eno ~\$90k; Galbot G1 ~\$91–97k; Keenon XMAN ~\$100k; PAL TIAGo Pro ~\$85–95k	Industrial/research purchase; the segment's center of gravity

Band	Examples (2026 street/list prices)	Model
>\$100k	Rainbow RB-Y1 commercial \$120k; Richtech ADAM \$180k; Ameca ~\$250k; CASBOT 02-class ¥328.8k; CloudMinds Ginger ¥698k; research heritage (PR2 \$400k)	Premium/entertainment/legacy
RaaS	ADAM \$3,500/robot/mo (25-unit 60-mo contract); AgiBot A2 ~\$20k+/mo; delivery-class \$300–800/mo; Eden \$10/hour; home \$449–499/mo (Weave, 1X NEO)	Labor-substitution pricing; dominant new-entrant model in the US

Note the China-vs-export gap: the same robot often costs 2–3× more through Western resellers (Atribot T1: ~\$14k domestic vs \$40,950 US reseller list) — tariffs, support and channel margins that an EU-based vendor does not pay locally. Pattern: Chinese street prices cluster at \$45–100k industrial (falling — some wheeled models below RMB 100k), Western/Korean at \$80–180k, and a new sub-\$15k consumer/edu front opened in 2025–26. Maintenance contracts (Deft: \$30k/yr on a \$40k robot) and software/support subscriptions (Galbot ~\$27k/robot/yr) shift lifetime value off the sticker price. History's warning: Pepper's ~\$9k consumer bundle lost money at ~27,000 units and the business died; unit economics, not unit counts, decide survival.

14 Safety, standards & regulation

- **ISO 13482** (personal care robots — mobile servant class fits semi-humanoids): 2014 edition current; revision (retitled toward "service robots", broadened to professional applications, FDIS stage mid-2025) expected imminently.
- **ISO 10218-1/-2:2025** published: industrial robot safety overhauled with functional safety, cybersecurity and collaborative requirements; TS 15066 folded in. **ANSI/RIA R15.08** covers industrial mobile robots with mounted manipulators; **UL 3300** (service robots in public spaces) joined OSHA's NRTL list Dec 2025.
- **The wheeled certification advantage:** statically stable platforms certify under all of the above *today*; dynamically balancing robots await **ISO 25785-1** (committee draft; expected 2026–27). IEEE's humanoid-standards pathway study (2025) puts ratified humanoid standards 18–36 months out.
- **EU:** Machinery Regulation 2023/1230 applies **20 January 2027** (no grace period): self-evolving-behaviour assessment, cybersecurity of safety functions, digital documentation — layered with the AI Act. Burden for laggards; moat for compliance-by-design vendors. TÜV-grade certification of a dual-arm mobile platform is an unclaimed European first.
- **USA — the 2026 discontinuity:** FCC Public Notice **DA 26-786** (28 July 2026) added "foreign-produced advanced robotic devices" to the Secure Networks Act Covered List: any mobile ground robot >4.4 lbs with environmental sensing, ≥200 kbps connectivity and AI autonomy loses FCC equipment authorization for new models unless granted a Department-of-War conditional approval. Fixed industrial robots are exempt; wheeled/tracked semi-humanoids are squarely in scope. Practical effects: Chinese vendors lose the US; Korean/Japanese/European vendors need approvals or US manufacturing ("domestic end product" per 48 CFR 25.101(a)); US integrators re-source components.
- **China:** MIIT stood up a humanoid/embodied-AI standardization committee (Nov 2025; Unitree and AgiBot founders as deputy chairs) and unveiled the first national humanoid standard system in March 2026; existing GB/T service-robot safety standards apply meanwhile. Shanghai issued the first humanoid governance guidelines (2024).

Observed reality: among 180 main-matrix robots, published robot-safety certifications are rare to absent — including for market leaders. Oversonic's "industrial + healthcare certified" claim resolves to CE-machinery + ISO 27001. A vendor that can put ISO 13482 / MR-2027-ready CE on a dual-arm mobile platform holds a genuinely scarce asset.

15 Deployment evidence: who is really working

Deployment class	Evidence (confidence)
Historic scale	Pepper: ~27,000 produced / ~17,000 sold, worldwide retail & reception (third-party) — still unmatched
Factory production shifts	AgiBot A2-W at Fulin Precision (~100-unit order; 4 robots moved 800+ boxes/3 h) and G2 at Longcheer (310 units/h tablet line) (third-party); Walden W0.5 in 8-h shifts at a Toyota plant since Feb 2026 (vendor/third-party); Midea MIRO U in its Wuxi plant (vendor); Zoomlion internal fleets (vendor)
Retail/pharmacy autonomy	Galbot: 10+ licensed Beijing robot pharmacies, ~170 kiosks in 40+ cities, 99.5% grasp claim over ~5,000 SKUs (vendor-claimed); Haohai "Juliet" pods on Nanjing Road (2,368 orders in first months, third-party)
Hospitality/F&B	Richtech ADAM: Walmart-hosted One Kitchen sites, Vegas café (16k+ drinks), arenas (vendor/third-party); Keenon XMAN-R1 at Shangri-La; Zerith H1 in 20+ hotels (third-party)
Healthcare/eldercare	Gary: ~10 units Philadelphia + Israeli MoH (third-party); Nyokkey at Fujita hospitals; Mirokai at AP-HP/Montpellier; Aeolus Ae0 in Japanese eldercare (operators are also investors)

Deployment class	Evidence (confidence)
Security/infrastructure	1X EVE via Everon/ADT: est. 150–250 US buildings (estimated); ugo Pro in Japanese buildings (third-party); Jinki Ittai on JR West railways — revenue service since July 2024 (third-party)
Logistics pilots	Reflex at GXO; Rainbow RB-Y1 at Coupang (confirmed); Dexterity Mech truck-loading pilots + Sanmina manufacturing (third-party)
Home (beta)	Weave Isaac deliveries from Sept 2026 (CA); Sunday Memo beta "by Thanksgiving 2026"; 1X NEO late 2026 — all pre-scale (vendor-claimed)

Contrast: flagship bipedal deployments remain single-to-low-double-digit robot counts (Figure/BMW: 2 robots; Agility/GXO: "small fleet"). The wheeled category's deployment lead is roughly two orders of magnitude.

16 Competitive assessment for a German entrant

Where the openings are

- **EU-certified industrial semi-humanoid.** No vendor holds robot-safety-grade certification for a dual-arm mobile platform under the incoming MR 2023/1230 regime. German TÜV culture + the domestic research bench (DLR torque control, KIT ARMAR manipulation, Franka-class compliant arms) make compliance-by-design the natural differentiator against spec-sheet-led Chinese imports.
- **Care & hospital logistics in Europe.** Demographics guarantee demand; the credible platforms are pilots (GARMI has no spin-off; HoLLiE stayed academic; MiPA is late; Mirokai is logistics-only). ISO 13482 + hospital-grade hygiene + Devanthro-style teleop fallback is an assemblable, defensible package.
- **US market via the FCC wall.** A German vendor with US final assembly ("domestic end product") can sell where Chinese rivals now cannot — the single largest regulatory gift to Western robot makers in years. Walden's \$1.1B day-one valuation prices exactly this scarcity.
- **Data flywheel as default architecture.** Ship every unit teleop-capable and log demonstrations under EU-grade data governance ("GDPR-clean embodied-AI data" is itself a product for European enterprise customers wary of Chinese and US clouds).
- **Component strategy.** Schaeffler (actuators; invested across Humanoid UK, Hexagon, reportedly Neura), igus (polymers), Franka/Agile (arms), SICK/IDS (sensing) form a domestic supply chain; the missing integrator role is the opportunity.

Threat tiers (analyst view)

Tier	Actors	Why
Severe	AgiBot, Galbot, UBTech; Keenon/Pudu channels; Unitree price disruption	Scale, capital, data, falling prices; EU entry inevitable (Rokae already exhibits in the West)
High	Toyota ELELY, Yaskawa(+Tokyo Robotics), Samsung/Rainbow, LG, Hexagon/Schaeffler, Walden	Incumbent capital + manufacturing + patient roadmaps
Medium	Humanoid UK, Neura, Sunday/Weave/Dexterity/Dyna, Richtech, Oversonic, Enchanted Tools	Funded and fast, but each thesis still unproven at scale
Watch	SwitchBot/consumer-price entrants; open-source kits; Genesis-class model labs; Vassar/Piggy/HABIT-type YC stealth	Price-floor and expectation-setting effects

Recommended posture

Enter as the certified European industrial/care semi-humanoid with a data flywheel. Concretely: (1) wheeled omnidirectional base + lift torso, dual 7-DoF compliant arms in the 5–10 kg/arm class, tool-changer + media at flange (an undisclosed axis across nearly all competitors); (2) MR-2027/ISO-13482 compliance engineered from day one and certified loudly; (3) launch vertical where ROI is provable within one budget cycle — intralogistics machine-tending or hospital logistics — with RaaS pricing against the \$3.5k/mo ADAM benchmark; (4) teleop + demonstration-data capture in every unit; (5) US strategy via local assembly partner from year one; (6) strategic component alliances (Schaeffler/Franka/igus) rather than full vertical integration. Speed matters: the Korean incumbent wave shows how fast listed industrials can pivot, and Chinese price curves compress the window in which "European + certified + trusted" commands a premium.

17 Annex A: Borderline & excluded systems

Documented for completeness; excluded from the main matrix. Full dossiers exist for each in the repository.

Robot / Company	Base	Classification note
Monty Anybots	two-wheel dynamically self-balancing	Reclassified to annex 2026-08-02 (user review): 2007 teleop research prototype, never productized — consistent treatment with othe
GoMate GAC Group	variable wheel-leg: 4-wheel mode (~1.4 m) / 2-wheel self-bal	GAC all-solid-state battery; claimed >80% energy reduction vs comparable humanoids. Third-generation GAC embodied robot. BORDERLIN
Stretch 3 Hello Robot	wheeled (differential drive) — BORDERLINE: single-arm, not a	BORDERLINE ENTRY (single arm) — include in borderline annex, not the main matrix. Added at user request as design-philosophy refer
AEON Hexagon Robotics (Hexagon AB)	BORDERLINE wheels-on-legs: articulated legs terminating in p	BORDERLINE morphology (wheels-on-legs) — flag when comparing against pure wheeled semi-humanoids. Strategic angle: metrology giant
EMIEW3 Hitachi	wheeled: drive wheels + small casters per foot; EMIEW lineag	Borderline inclusion: no manipulation capability. EMIEW = Excellent Mobility and Interactive Existence as Workmate. Original 2005
L1 Moon Dynamics	two-wheel dynamic-balancing wheel-LEG hybrid (BORDERLINE sco	BORDERLINE scope: wheel-leg (wheeled-bipedal) robot, included as boundary datapoint like Hexagon AEON / GAC GoMate. Company extrem
Moby Noble Machines (ex-Under Control Robotics)	BIPEDAL/legged (NOT wheeled - contradicts discovery classifi	SCOPE: bipedal legged humanoid, not a wheeled semi-humanoid - discovery note '9kg lift, wheeled' was wrong on both counts (payload
Bot Handy Samsung Research (Samsung Electronics)	wheeled cylindrical base (drive type not disclosed); single-	Borderline: single-arm concept, included for Samsung-strategy context. Samsung's real semi-humanoid position runs via 35% subsidia
The Five (XiaoWu) Tencent Robotics X	wheel-leg hybrid: four straight telescoping legs with wheels	BORDERLINE for study scope: wheel-leg hybrid (4 straight telescoping legs with wheels), not a wheeled-base semi-humanoid. Fifth-ge
Human Support Robot (HSR) Toyota	wheeled (omnidirectional) — BORDERLINE: single-arm, not a se	BORDERLINE ENTRY (single arm) — borderline annex, not main matrix. Added at user request as the historic home-assistance archetype
Calvin-40 Wandercraft	BIPEDAL (two legs, headless) — EXCLUDED from semi-humanoid s	EXCLUDED-as-biped: kept in study for Renault deployment context. Survey-sweep 'wheeled' claim was wrong; startup-sweep 'biped' cla

18 Annex B: Research repository guide

This study compresses a research repository delivered alongside it:

- **companies/<company>/company.md** — company profile: ownership, funding, strategy, sources.
- **companies/<company>/<robot>/robot.md** — full dossier: morphology, locomotion, manipulation, sensing, actuation, compute/software, safety, deployment evidence, analyst assessment, numbered sources.
- **.../specs.json** — machine-readable spec sheet; every field carries value + confidence (vendor-claimed / third-party / estimated) + source URL. These files generate the xlsx.
- **.../images/** — downloaded imagery, each visually verified; images.md maps every file to its source page, direct URL, caption and licence note.

- **semi_humanoid_market_2026.xlsx** — Overview, full 34-column spec matrix, per-region sheets, Borderline & excluded, ~3,400-row per-field source register, confidence colour legend.

Compiled 2 August 2026. Sources were accessed August 2026; the per-field register in the xlsx is authoritative for citations. Confidence discipline: vendor-claimed figures are marketing until deployed; third-party validation is noted wherever it exists; conflicts are preserved, not averaged.